

A Conceptual Study of Mobile Differential GNSS Architecture Utilizing UAV Networks

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ABSTRACT

Differential Global Navigation Satellite Systems (DGNSS) which feature rapid deployment and mobility can provide required navigation accuracy and integrity to a wide range of users at arbitrary locations where no permanently installed systems exist. This paper proposes a concept of mobile DGNSS whose reference stations are composed of multiple Unmanned Aerial Vehicles (UAVs) to facilitate mobility of system. Key algorithms of Mobile DGNSS architecture, developed in this work, include a decision making technical for selecting reference station location, a fast location surveying method and integrity monitors.

Using the decision making technical, landing positions of UAV reference stations are selected by taking into account the geographical characteristics of the landing vicinity, and the distances from DGNSS service demands. To validate that the locations satisfy the minimum siting requirements upon landing, two tests are carried out: territorial inclination test for assuring a sufficient number of satellites in view and satellite visibility test for avoiding a harsh environment with obstacles. The fast location surveying method is developed to reduce the time taken for the determination of the surveyed positions within specific positional accuracy. The simulation results show that the proposed method enables users to achieve the desired positional accuracy in less amount of time than the general method of simply averaging stand-alone GNSS solutions only. Integrity monitors in addition to the well-established Ground Based Augmentation System (GBAS) monitors are developed to detect faults specific to the mobile DGNSS. The code multipath monitor and the station displacement monitor are designed to detect large multipath errors on code measurements and unexpected movements of the reference stations, respectively under uncontrolled siting conditions.

Rapid start-up time is a critical factor to assess whether the proposed concept is feasible. We derive surveying position error budgets for different multipath environments and system architectures. A new Vertical Protection Level (VPL), which includes the bound of surveying errors, is defined for the mobile DGNSS users. The effect of the resulting surveying errors on the system is evaluated by estimating deployment time for various conditions.

1.0 INTRODUCTION

Unmanned vehicles are currently used for a number of missions including reconnaissance, surveillance, and combat. One of the most essential technologies required for the operation of unmanned systems is reliable and accurate computation of navigation information. As Global Satellite Navigation System (GNSS) is an essential component of most autonomous navigation systems, it is imperative to research a way to improve the accuracy and the reliability of GNSS positioning. In order to achieve the high accuracy and reliability that is needed for the autonomous Unmanned Aerial Vehicle (UAV) applications, Ground Based Augmentation System (GBAS) technology, which was developed to support precision approach and landing of manned aircraft, can be employed [1]. GBAS ground facility is installed at fixed positions within the property of a particular airport, and the facilities broadcast differential corrections and provide integrity information to users. However the range of the fixed reference stations are limited to within several tens of kilometers of GBAS equipped airports. If the reference stations can be deployed at arbitrary mission locations where GBAS service was not originally available, the constraint of the correction broadcasting range is no longer a limiting factor. Furthermore, if the reference stations are mobile, even if the mission locations change, there is no need for additional GBAS installations, which implies that the overall installation costs can be reduced.

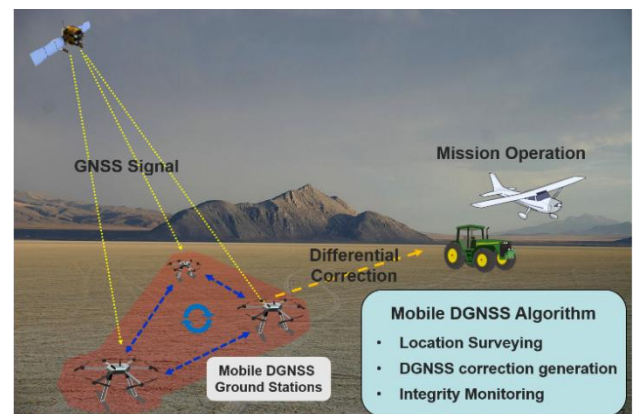


Figure 1 Mobile DGNSS Concept

Fig. 1 illustrates the concept of mobile DGNSS for supporting a wide variety of users by providing navigation capabilities of high accuracy and integrity. The

mobile DGNSS is composed of three or more UAVs equipped with a GNSS receiver and a RF data broadcast transmitter.

As the reference stations for this system are UAVs, the system does not have to be geographically fixed, can be deployed at arbitrary locations of interest, and additional UAVs can be incorporated to extend the range of DGNSS service. Furthermore, the system monitors any system failures which may pose integrity risks to the users, and provides differential corrections and integrity information to GNSS users who are carrying out their missions. In order to establish this system, various conditions must be met first. The reference stations should be deployed at appropriate locations where they can receive GNSS signals without severe disturbances. Also, they should be able to provide DGNSS service to the users which are carrying out the missions. Additionally, the arrangement of the stations should be spread out in order to reduce the correlation of multipath errors between each receiver. After the deployment of mobile DGNSS, the reference stations should be able to determine their accurate surveyed positions in a short period of time. This is because inaccurate surveyed locations of the reference stations increase differential correction errors and degrade position solutions calculated by the DGNSS user.

This paper proposes a concept of mobile DGNSS architecture which utilizes multiple UAVs as ground reference stations to provide navigation capabilities of high accuracy and integrity to users at mission locations. We introduce essential techniques to address key issues for the realization of this system. Section 2 develops a decision making technique to select landing points for the mobile DGNSS reference stations considering the requirements of the system and the inherent limitations of the mission locations. As the deployment locations are not in restricted areas such as airports, the DGNSS reference stations may be significantly affected by the multipath effects from the ground and obstacle reflections, territorial inclinations and satellite blockages. In Section 3, a methodology for the fast location surveying of the reference stations is proposed to reduce the deployment time for the initiation of DGNSS service. Section 4 introduces integrity monitors developed to detect faults specific to the mobile DGNSS. In Section 5, the feasibility of the proposed mobile DGNSS system are assessed by deriving surveying position error budgets and estimating system deployment time. This paper is concluded in Section 6.

2.0 SELECTION OF REFERENCE STATION LOCATIONS

When three or more UAVs are deployed at arbitrary mission locations to serve as reference stations, they have to fulfill the demands for DGNSS operation while taking into account various environmental factors. First of all, the UAVs find a central point where the sum of the

distances from each DGNSS demand is minimal. This is shown in Eq. (1).

$$\text{Min} \sum_i D_i d_i \quad (1)$$

$$D_i \in \{0,1\} \quad (i = 1,2, \dots, N)$$

where D_i is the i^{th} DGNSS demand and d_i is the distance between the point and D_i location.

After computing the point, the UAVs fly to, find and land at appropriate locations around the point where the terrains are vacant and flat. The process of finding suitable terrains is carried out using image sensors onboard the UAVs. Upon landing, the arrangements of the UAVs should be spread out in order to reduce the correlation of multipath errors between each UAV receiver.

After landing, the UAVs (i.e., reference stations) assess their environments using requirement tests. The tests consist of two parts: territorial inclination test and satellite visibility test. The territorial inclination test measures the UAVs' surrounding terrain inclination using their onboard accelerometer. This test is necessary as large amounts of inclination on the airframe (i.e. GNSS antenna) degrade the antenna performance and reduce the number of visible satellites. Generally an elevation cutoff angle of 5 degrees is used (At lower cutoff angles, strength of the received signals is weaker, experienced multipath errors are greater, and there is higher probability of loss of lock.). Consequently for the test, we set a threshold value of 5 degrees for the inclination to prevent the lack of satellite visibility. If the measured inclination exceeds the threshold of 5 degrees, we consider that a failure has arisen. The satellite visibility test verifies that the visible satellite signals are received without significant loss of signal. The test compares the number of received satellite signals to the number of possibly visible satellites which is computed from the broadcast ephemeris message. The discrepancy of these two numbers indicates satellite blockage. Therefore, if the difference exceeds a pre-determined threshold, the UAV searches for a new location.

3.0 FAST LOCATION SURVEYING METHOD

Once the UAVs land at the optimal landing sites, the greatest limiting factor for the initiation of mobile DGNSS service is the time taken for the surveyed locations of assured level of accuracy to be determined. Therefore, in order to reduce the overall deployment time until operation, a methodology for fast location surveying of the mobile reference stations should be developed.

3.1 ALGORITHM

In this paper, we adapted a methodology proposed by Schultz **요류! 참조 원본을 찾을 수 없습니다.** which

uses accurate relative distances to correct inaccurate GNSS-based positions. This methodology requires one prerequisite and two steps. The prerequisite is the use of accurate relative distances between the reference stations. The first step is the formation of an accurate geometry using the relative distances. Then, in the second step an optimal rigid body transformation is carried out on the geometry. Previous work (summarized in Subsection 3.1.1) showed that precise relative distances can be obtained using GNSS carrier phase measurements 오류!

참조 원본을 찾을 수 없습니다.. In Subsection 3.1.2, we improved computational speed by utilizing relative position vectors (instead of absolute relative distances) to form the accurate geometry. The singular value decomposition computation is used to find the improved position solution which minimizes an objective function, as described in Subsection 3.1.3. Overall, the methodology plays the role of accurate repositioning of standalone GNSS position solutions. By averaging these improved position estimates (rather than the raw solutions) over time, the surveyed positions can be determined in a shorter period of time.

3.1.1 Accurate Relative Distance Estimation Method

There are several options for obtaining accurate relative distances between UAVs. For example, lidar or radar sensors provide relative distance measurements of high accuracy. In addition, carrier-phase measurements obtained from GNSS receivers can also be used to determine accurate relative distances. Compared to lidar or radar sensors, if only carrier-phase measurements are used, some limitations such as the need for direct line of sight between reference stations and the need for additional onboard sensor equipment can be alleviated. Therefore we adapted the relative position computation method developed for the Joint Precision Approach and Landing System (JPALS) 오류! 참조 원본을 찾을 수

없습니다.. The method with which a centimeter-level relative positioning accuracy is achievable is summarized as below. The relative distance is estimated by employing the double difference of carrier-phase and widelane geometry-free observables. This process firstly computes the widelane geometry-free observables using dual frequency code and carrier measurements, and filters the observables with a moving-average filter. After the integers of the filtered widelane observables are estimated, the double difference of carrier phase and widelane geometry-free observables are calculated. As the reference stations of mobile DGNSS are not too far away from one another, the remaining ionospheric and tropospheric delays of the double difference observables are negligibly small. This helps to solve for carrier-phase integer ambiguity. The relative positions and carrier-phase integer differences are estimated using the least-square method. However, as the least square method can only provide floating solution of integers, an additional process is required to constrain the ambiguities as integers.

To fix the integers, the well-known Least-squares Ambiguity Decorrelation Adjustment (LAMBDA) method is applied 오류! 참조 원본을 찾을 수 없습니다. 오류! 참조 원본을 찾을 수 없습니다..

3.1.2 Geometry Formation

After the accurate relative distances are determined, we move onto the formation of an accurate geometry using the relative vectors. As the carrier-phase relative position method provides vector positions between the reference stations, we form a geometry consisting of the point set $(\vec{A}_{1,g}, \vec{B}_{1,g}, \vec{C}_{1,g})$ where $\vec{A}_{1,g} = (x_{A1,g}, y_{A1,g}, z_{A1,g})^T$, $\vec{B}_{1,g} = (x_{B1,g}, y_{B1,g}, z_{B1,g})^T$, $\vec{C}_{1,g} = (x_{C1,g}, y_{C1,g}, z_{C1,g})^T$ using one of the GNSS receivers and the relative vectors from that receiver. For example, if positions $(\vec{A}_1, \vec{B}_1, \vec{C}_1)$ are the GNSS positions and we use the receiver at position $\vec{C}_1$ for the formation of the geometry:

$$\begin{aligned} & \left[\vec{A}_{1,g}, \vec{B}_{1,g}, \vec{C}_{1,g} \right] \\ &= \left[\vec{C}_1 + \vec{L}_{CA}, \vec{C}_1 + \vec{L}_{CB}, \vec{C}_1 \right] \end{aligned} \quad (2)$$

where $\vec{L}_{CB}$ is the relative vector from true positions C to B and $\vec{L}_{CA}$ is the relative vector from true positions C to A , both of which are found from the algorithm described in Subsection 3.1.1.

In the method proposed by Schultz 오류! 참조 원본을 찾을 수 없습니다., the geometry is found by fixing two points and moving one point at a time so that two of the accurate relative distance magnitudes are satisfied at a time. This requires iterations to obtain a geometry that satisfactorily meets the relative distance requirements. However, if relative distance vectors rather than scalar ones are utilized as shown in Eq. (2), iterations are not required and the solution can be obtained in a one step process. Consequently, less computation load is required, which leads to more efficient implementation of the algorithm for real time systems. Furthermore, if we increase the number of reference stations for this step to form a geometry of degree greater than three, the benefit of using vector information would become greater. An illustration of the geometry formation is shown in Fig. 2.

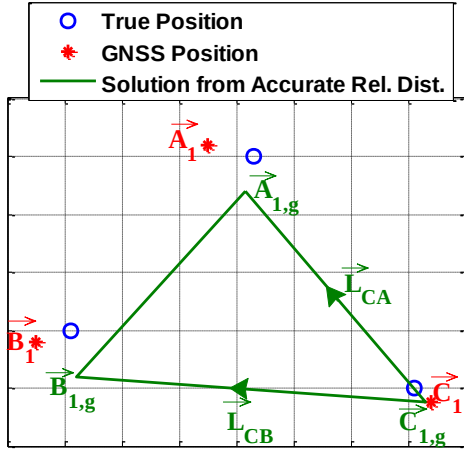


Figure 2 Illustration of Geometry Formation

3.1.3 Repositioning via Rigid Body Transformation

After the formation of the geometry, we follow the architecture outlined by Schultz 오류! 참조 원본을 찾을 수 없습니다. and carry out translation and rotation on the geometry in the Euclidean space to find the positions $(\vec{A}', \vec{B}', \vec{C}')$ such that the sum of error squares between the GNSS positions and the vertices of the geometry is minimized. In other words, we minimize a cost function, J .

$$J = \sqrt{\vec{\varepsilon}^T \vec{\varepsilon}} \quad (3)$$

where $\vec{\varepsilon}$ is the 3 by 1 error vector whose components are the errors between the GNSS positions $(\vec{A}_1, \vec{B}_1, \vec{C}_1)$ and the final results $(\vec{A}', \vec{B}', \vec{C}')$. Instead of using the quaternion method proposed by Schultz 오류! 참조 원본을 찾을 수 없습니다., the computation is carried out using the Singular Value Decomposition (SVD) algorithm. The advantage of using the SVD approach to perform least-square geometric alignment is that this generalizes to any n-dimensional applications where n is greater than three as shown in 오류! 참조 원본을 찾을 수 없습니다..

First, the centroids of the geometry and the GNSS positions, $\vec{P}_{cent,geo}$ and $\vec{P}_{cent,GNSS}$, are computed respectively. Then, the rotation matrix, R , between the two dataset is calculated using singular value decomposition.

$$H = (\vec{A}_{1,g} - \vec{P}_{cent,geo})(\vec{A}_1 - \vec{P}_{cent,GNSS})^T + (\vec{B}_{1,g} - \vec{P}_{cent,geo})(\vec{B}_1 - \vec{P}_{cent,GNSS})^T + (\vec{C}_{1,g} - \vec{P}_{cent,geo})(\vec{C}_1 - \vec{P}_{cent,GNSS})^T \quad (4)$$

$$[U, S, V] = SVD(H) \quad (5)$$

$$R = VU^T \quad (6)$$

where H is number of points multiplied by the cross covariance matrix of geometry dataset and GNSS positions dataset, U is the unitary matrix, S is the diagonal matrix, and V is the conjugate transpose of a unitary matrix. Finally, the translation vector, $\vec{t}$ and the optimized solution $(\vec{A}', \vec{B}', \vec{C}')$ are computed.

$$\vec{t} = -R \times \vec{P}_{cent,geo} + \vec{P}_{cent,GNSS} \quad (7)$$

$$\begin{bmatrix} \vec{A}' & \vec{B}' & \vec{C}' \end{bmatrix} = R * \begin{bmatrix} \vec{A}_{1,g} & \vec{B}_{1,g} & \vec{C}_{1,g} \end{bmatrix} + \begin{bmatrix} \vec{t} & \vec{t} & \vec{t} \end{bmatrix} \quad (8)$$

An illustration of this step is shown in Fig. 3. This algorithm is expected to return the processed positions which are closer to the true positions than the GNSS positions. By utilizing more accurate positions to determine surveyed positions of reference stations, we can reduce deployment time required prior to initiating an operation of the mobile DGNSS system.

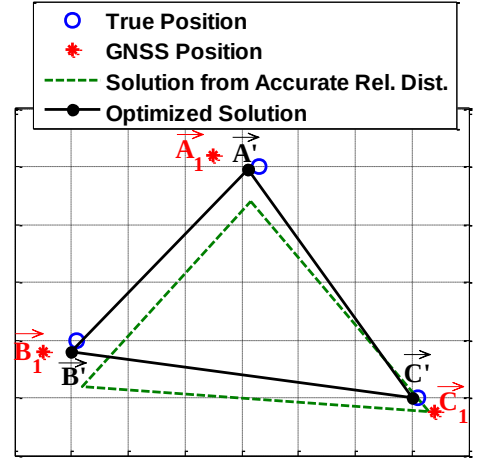


Figure 3 Illustration of Repositioning to Solve Optimized Solution

3.2 TEST AND SIMULATION RESULTS

Using the data obtained from KAIST Local Area Differential-GNSS (LAD-GNSS) integrity monitor test-bed, the accurate relative distance estimation step of fast surveying method was validated. The details on the LAD-GNSS test-bed are described in Section 5. Two base stations which are separated by 46 m were selected and their relative distances were estimated using L1/L2 code and carrier phase range data. Fig. 4 shows the accuracy of the estimated relative distances between the two reference stations. The relative distance errors were found to be completely bounded within the ± 2 cm level for the entire 40 hours period. The two base stations are relatively close from one another, which make the remaining errors (i.e., ionospheric delay and tropospheric delay) after double differencing almost negligible. This helps us to estimate carrier-phase integers accurately and consequentially to calculate the relative distances with centimeter-level

accuracy as shown in Fig. 4. Also, if we compare this accuracy to that of standalone GNSS positioning errors, we can confirm that the relative distance errors are considerably less significant, which is the condition required for the following two steps.

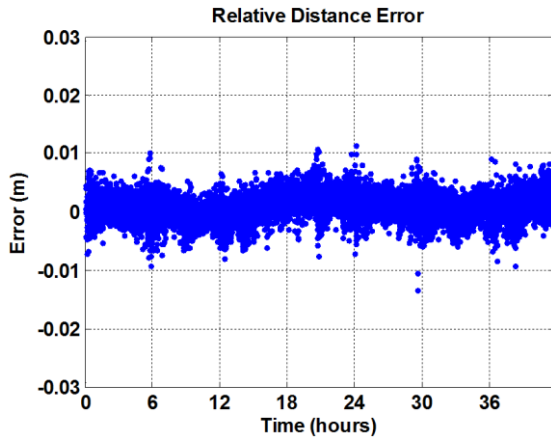


Figure 4 Relative Distance Error between Two Stations

To validate that the fast surveying method is effective in lowering the overall position error, a simulation was carried out. To create the simulation data, for a given set of true positions, random errors of zero mean Gaussian distribution with a standard deviation of 5 m were injected into each of the coordinates. The error-injected positions were assumed to be GNSS positions. Assuming that the receivers use a code-carrier smoothing filter with a 100-seconds window to compute GNSS positions, the GNSS positions for the simulation were created with 100 seconds intervals to take into account the independent sampling. Fig. 5 shows the errors between the surveyed positions and the true positions for each station, and the total error reductions from the fast surveying method. The top three graphs, red, green and black lines, represent the errors for the individual stations. The dotted curves are the surveying position errors which are computed by differentiating the error-injected positions (i.e., simulated standalone GNSS solutions) and the true positions. As shown in the Fig.5, the surveyed position errors decrease as time goes on due to the averaging effect of errors. If the fast surveying method is applied, the errors of the standalone solutions decrease at each epoch. Consequently after applying the method, the surveying errors, which are represented as solid curves in Fig. 5, decrease faster than the dotted curves. Note that different values were added to all of the data in order to make the results visibly clear. Note that the added numbers (indicated in the legend) have no particular significance.

The blue curve in Fig. 5 is the sum of error reductions of all stations at each epoch after applying the algorithm. The error reduction is obtained by subtracting surveying errors after processing from surveying errors before processing. As shown in the graph, the error reductions are up to 2 m in the first two hours, but the reduction is

reduced over time. The reason is that, as the positions are averaged over time, their overall errors are also decreased. Despite the increase in the accuracy for discrete GNSS position solutions by the algorithm, if the overall error is low, the differences in the accuracies between the pre-processed and post-processed data are less evident. The results suggest that the processed positions allow the user to obtain accurate surveyed positions in a shorter amount of time than raw GNSS data. In the future, to assess the effectiveness of the fast location survey algorithm in realistic situations, we will test the performance of this algorithm using the GNSS data collected from the mobile DGNSS prototype.

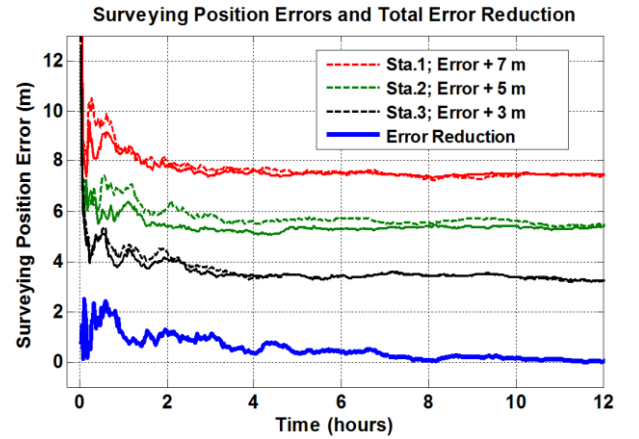


Figure 5 Comparison of Surveying Position Errors and Total Error Reduction from Fast Surveying Method

4.0 INTEGRITY MONITORING

Mobile DGNSS operates in unpredictable environments where the terrain may be inclined or the surrounding forestry may be dense. Moreover, reference stations can be unexpectedly moved by external disturbance such as a strong wind or a hit by a moving object. As uncontrolled siting conditions may cause inaccurate differential corrections, additional monitors are needed to detect those faults. In this paper, we propose two monitors to detect severe multipath and unexpected displacement of reference stations, respectively. The most effective one to detect large multipath errors among the existing GBAS monitors is the B-value check which measures pseudo-range correction differences across reference receivers. However, the multipath error correlation between individual reference receivers can be large because mobile DGNSS reference stations may not meet the siting criteria equivalent to those of GBAS. Multiple reference stations may experience a sudden increase of multipath simultaneously at uncontrolled sites. If these happen, the B-value monitor would not perform as well as originally designed. Thus, we add the code multipath monitor to detect large multipath errors on code measurements.

A test statistic, MP , is generated based on a linear combination of L1/L2 code (ρ_{L1} , ρ_{L2}) and carrier-phase

(ϕ_{L1}, ϕ_{L2}) measurements. The linear combination is defined as [8]:

$$MP \equiv \rho_{L1} - \left(1 + \frac{2}{\gamma - 1}\right) \phi_{L1} + \left(\frac{2}{\gamma - 1}\right) \phi_{L2} \quad (9)$$

$$= M_{L1} + B - \left(1 + \frac{2}{\gamma - 1}\right) m_{L1} + \left(\frac{2}{\gamma - 1}\right) m_{L2} + \varepsilon$$

$$\gamma = \frac{f_{L1}^2}{f_{L2}^2} \quad (10)$$

M_{Li} and m_{Li} are the multipath errors on code and carrier-phase measurements on the Li ($i = 1, 2$) signals, respectively. ε is receiver noise and γ is the square of the frequency ratio. The bias term, B is:

$$B \equiv -\left(1 + \frac{2}{\gamma - 1}\right) N_{L1} + \left(\frac{2}{\gamma - 1}\right) N_{L2} \quad (11)$$

N_{Li} is the integer ambiguity of the Li frequency signals. The bias, B of MP is assumed to be a constant unless there is a cycle slip in the tracking of $L1$ and $L2$. Therefore, the constant is eliminated by subtracting the average value of MP . Although the portion of carrier-phase multipath is included in MP values, the amount is small compared to that of code multipath. Thus, the bias-removed MP can be approximated to be the multipath errors on $L1$ code measurements.

The code multipath monitor calculates the test statistic and compares the value of the test statistic to a corresponding threshold in order to determine if there is a failure. The threshold is defined using a general threshold determination method utilized by GBAS monitors [9]. In this method, GNSS data in a nominal condition are collected to calculate the test statistic. Next, for each of the nine elevation bins of 10 degrees, the MP sample standard deviation, $\sigma_{MP, elev}$, and MP sample mean, μ_{elev} , are calculated. To cover the non-Gaussian tails of the actual distribution, the standard deviation inflation factor, f , is estimated through the overbounding process [10]. Finally, the elevation-dependent threshold is modeled by Eq. (12).

$$MP \text{ Threshold}_{elev} = \mu_{elev} \pm Kf\sigma_{MP, elev} \quad (12)$$

($elev = 10, 20, \dots, 90$ degrees)

where, the K -factor is set to provide a fault-free alarm rate required for the system [11].

To determine the threshold of code multipath monitor, the dual-frequency data were collected from LAD-GNSS integrity monitor test-bed and the multipath errors on code measurements were estimated using the algorithm described above. In Fig. 6, the blue dots represent the multipath errors observed from the three reference receivers at various elevation angles. The red curve indicates the detection thresholds. The x-axis represents the elevation angles of satellites, while the y-axis represents multipath errors in meters. As shown in Fig. 7,

to determine the thresholds, we normalize the multipath errors and through the overbounding process, estimate the inflation factor, f , as 2.71. Using the computed inflation factor and the K -factor of six, the threshold is determined through Eq. (12). As shown in Fig. 6, we can see that the multipath estimates stay well below the detection threshold.

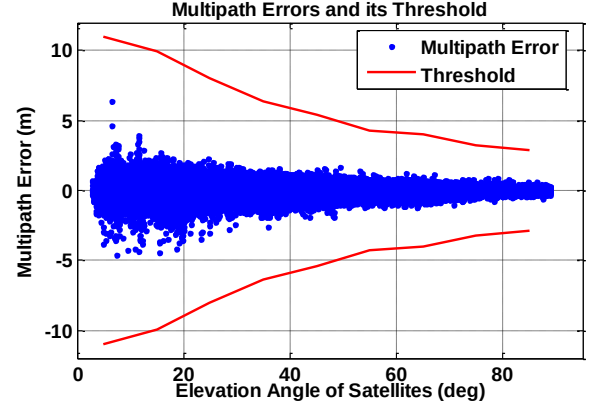


Figure 6 Multipath Errors on Code Measurement and its Threshold

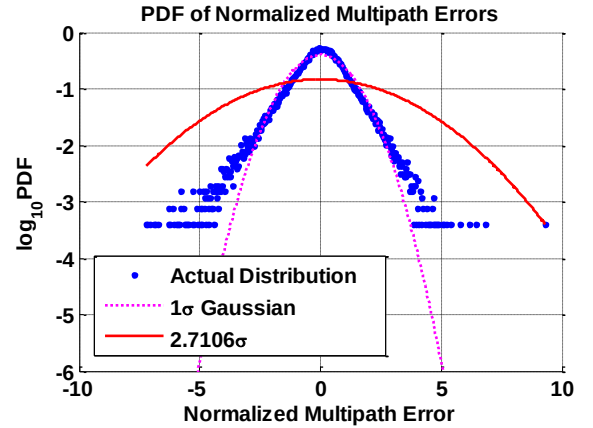


Figure 7 PDF of Normalized Multipath Errors and Sigma Inflation

After the locations of reference stations are surveyed, the master station broadcast differential GNSS corrections and integrity information to users that carry out missions. The mobile DGNSS includes LAD-GNSS integrity monitors which were evolved from GBAS monitors [1]. However, unlike GBAS or LAD-GNSS installations, the mobile reference stations are not fixed at their landing positions. While broadcasting differential corrections, external influences such as strong gusts of wind can unexpectedly move the reference stations. In this paper, we propose a station displacement monitor to detect the possible unexpected movement of reference stations. This monitor uses relative positions between the reference stations to verify that the stations stand still. Relative position vectors, $\mathbf{X}_{Rel}$, are computed through the accurate relative distance estimation method described in Subsection 3.1.1. In this test if the difference between the

relative position of receiver i with respect to receiver, $\mathbf{X}_{Rel,j}^i(k)$, at epoch k and the relative position, $\mathbf{X}_{Rel,j}^i(k-1)$, at epoch $k-1$ exceeds a threshold, the monitor indicates a failure and the surveyed locations of reference stations should be determined again.

$$|\mathbf{X}_{Rel,j}^i(k) - \mathbf{X}_{Rel,j}^i(k-1)| > Threshold \quad (13)$$

The threshold can be defined in the same manner as the code multipath monitor. However, in this case, the threshold is a constant and not an elevation-dependent function as shown in Eq. (14).

$$Threshold = \mu \pm f k \sigma \quad (14)$$

where, the K -factor level is set to meet a required fault-free alarm rate which is to be determined for a specific application of this system. Future work will derive a threshold of the proposed monitor based on test statistics collected from a mobile DGNSS prototype and perform both nominal and failure tests.

5.0 MOBILE DGNSS FEASIBILITY

5.1 SURVEYING POSITION ERROR BUDGETS

The time taken for the surveyed positions to be determined is the critical factor that will determine the performance and the feasibility of the proposed mobile DGNSS. As shown in the simulation result in Fig. 5, the benefit of the fast location surveying algorithm is apparent. However, the actual size of the gain which would be obtained from the algorithm differs depending on the GNSS error conditions. Thus, more work is needed to understand and then bound the degree of surveying error reduction. In this paper, we investigate surveying position error budgets without taking into account the effects of the fast surveying method. Also note that we make no use of the conventional Real Time Kinematic (RTK) techniques for self-surveying of the proposed system. The surveying position error budgets in this section are based on the use of carrier-smoothed-code measurements only. However, a more comprehensive analysis of the effectiveness of the fast location survey algorithm will be addressed in subsequent research.



Figure 8 KAIST LAD-GNSS Test-bed Hardware Configuration

Fig. 8 shows the KAIST Local Area Differential-GNSS (LAD-GNSS) Test-bed, which is aimed at meeting GBAS Category I (CAT-I) integrity performance. Using this test-bed, we first developed a surveying position error budget. As the multipath levels for static, rooftop-sited antennas of this test-bed are similar to that of the GBAS GAD-B error model 오류! 참조 원본을 찾을 수 없습니다., the GAD-B model is used for the multipath environment at this site. Using the error model and the KAIST LAD-GNSS system architecture, we carried out a Monte Carlo Simulation to obtain position error estimates at each simulation time.

In particular, our approach for the derivation of the position error estimates assumes that the pseudorange measurement errors are uncorrelated and have a common variance. We generated single snapshots of Vertical Dilution of Precision (VDOP) randomly from the probability distribution that is predefined based on VDOP observations of the KAIST LAD-GNSS architecture over one day. We then applied the pseudorange error model to determine a snapshot of the RMS ranging error. Using the simulated VDOP and the ground system error model, we obtained 95% position error estimates and derived a self-survey error budget by averaging these error estimates over time. Note that, in the simulation, the multipath error for the self-survey error is obtained from the GAD-B model, while the other ranging error budgets which include satellite clock and ephemeris errors and atmospheric propagation modeling errors (i.e., ionospheric and tropospheric errors) come from that of the L1 signal 오류! 참조 원본을 찾을 수 없습니다.. The simulation result is shown in Fig. 9.

In Fig. 9, the red line represents the change in 95% surveying error over the 24-hour period obtained from Monte Carlo Simulation. Since the surveyed positions are found by averaging the position estimates over time independently for each receiver, the surveyed position errors decrease as time goes on. In this averaging procedure, we generated position error estimates every 100 seconds to simulate independent errors considering that the actual system applies a code-carrier smoothing

filter with a smoothing time constant of 100 seconds. The black lines show the surveyed position errors obtained from data collected from three KAIST LAD-GNSS ground receivers. As we can see in the figure, the pre-specified error budget well describes the actual data trend.

Self-survey performance for mobile DGNSS can widely vary for different multipath environments and system architectures. Therefore, we carried out a surveyed position error budget simulation for different error conditions. As for the previous simulation, the randomly generated VDOP values and measurement error budgets for various conditions are used as the input of the surveyed error budget simulation. Fig. 10 shows the simulation results for vertical surveyed position error. The green curve represents the error budget assuming that the multipath level is worse than that of GAD-A, which was defined to reflect a level of performance achievable with low-cost GBAS installations 오류! 참조 원본을 찾을 수

없습니다.. Thus the vertical surveyed position error is greater than the baseline condition in red. The yellow line shows the error budget when multipath and noise error on L5 is assumed, and L5 is used in place of L1. For simulation purposes, the multipath and noise level for L5 were determined to be approximately 0.8 times that of GAD-B. This approximation is based on the graphical result of σ_{pr_gnd} value that is evaluated based on the data collected from the four Block IIF GPS satellites as

presented in 오류! 참조 원본을 찾을 수 없습니다.. As shown in the figure, the reduction by a factor of 0.8 may not be very critical to the resulting surveying position error budget. This is because the contribution of ground multipath and noise error to the total ranging error budget for each receiver is not very significant compared to the other error sources including the ionosphere.

The orange curve shows the error budget when a dual-frequency system with code-carrier divergence free smoothing is applied. In this case, to see how much improvement can be achievable by using the code-carrier divergence free smoothing technique, we assume that the ranging error variance associated with ground multipath is approximately close to zero. However, despite the improvement in multipath error, the influence of other error sources, such as ionospheric error, is dominant; thus the surveyed position error improvement is only minor. In the case of a dual-frequency system using ionosphere-free measurements, shown in blue, the noise and multipath in the ionosphere-free measurement is about 2.6 times larger than that of single-frequency code measurements 오류!

참조 원본을 찾을 수 없습니다.. However, as the ionosphere-induced error on the standalone ranging signal for each reference receiver is removed, a significant improvement can be notice

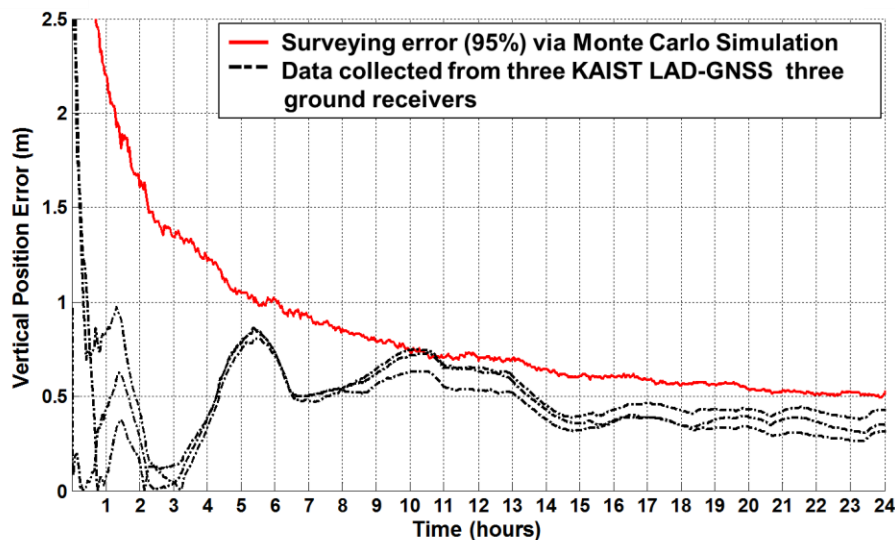


Figure 9 Vertical surveying position error budget obtained by Monte Carlo Simulation based on the multipath condition and based on system architecture of KAIST LAD-GNSS (red line) and the surveying position error obtained from KAIST LAD-GNSS receiver data (black line)

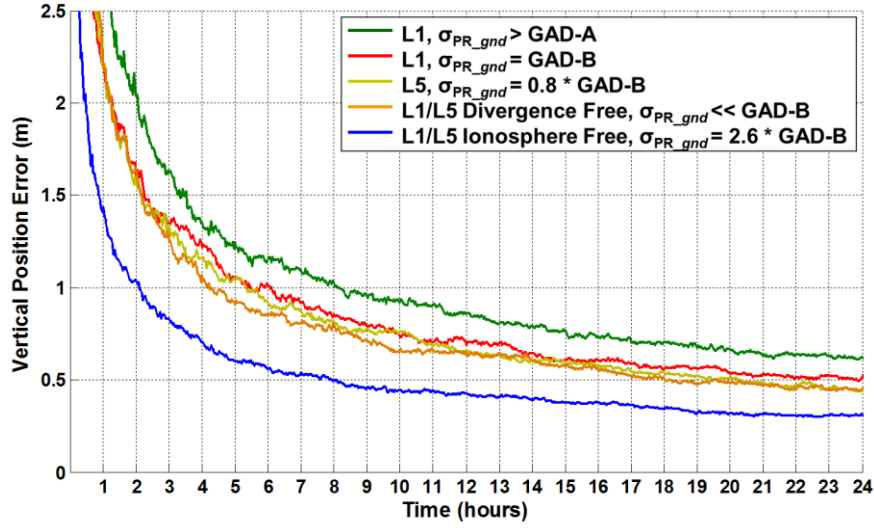


Figure 40 Vertical surveying position error budgets depending on different multipath conditions and system architectures

5.2 EFFECT OF SURVEYING ERROR ON SYSTEM

Unlike most GBAS systems, correction performance of mobile DGNSS is significantly affected by the surveyed position errors of reference stations. Therefore, using the surveyed position error budgets derived in this paper (see Fig. 10), we investigate the effects of the surveyed position error on system performance. For this, we define the Vertical Protection Level (VPL) for the mobile DGNSS, VPL_{mobile} , by combining surveying position error budgets with VPL for GBAS, VPL_{GBAS} , which is computed as defined in the RTCA LAAS MOPS 오류!

참조 원본을 찾을 수 없습니다.:

$$VPL_{mobile} = VPL_{GBAS} + K\sigma_{vert,surveying\ error} \quad (15)$$

where $\sigma_{vert,surveying\ error}$ is the RMS surveying error in the position domain which is derived from the surveying position error budget, and K is an inflation factor that propagates the surveying error variance to a level of 10^{-7} integrity using a zero-mean Gaussian distribution (note that this requires that the zero-mean Gaussian distribution with a standard deviation of $\sigma_{vert,surveying\ error}$ bounds the actual error distribution to this probability).

In Fig. 11, the red line shows the value of VPL_{mobile} as computed by Eq. (15), and the magenta line represents VPL_{GBAS} alone. $\sigma_{vert,surveying\ error}$ in Eq. (15) is computed by dividing the values of the red curve in Fig. 10 by two, since the red curve shows the baseline surveyed position error budget in 95 % (i.e., two times $\sigma_{vert,surveying\ error}$). For the VPL_{mobile} computation in Fig. 11, the standard RTCA 24-satellite GPS constellation 오류! 참조 원본을 찾을 수 없습니다. is used, and the mobile DGNSS user is assumed to be located at the KAIST campus. As the surveyed position error budget decreases over time, VPL_{mobile} also has a decreasing trend. As shown in Fig. 11, during the first hour or so after surveying begins, VPL_{mobile} exceeds the Vertical Alert

Limit (VAL) for CAT-I service (i.e., 10 meters) which is represented as a black line, so the system is not available at the CAT-I decision height of 200 ft, although it might be usable for the higher decision heights (250 – 400 ft) supported by LPV approaches.

The starting time of system operation may be at any point in a 24 hour period. Therefore, to determine the most conservative deployment time, the maximum mobile VPL value over the entire period should be selected. For each deployment time, the maximum mobile VPL over 24 hours for each error condition is plotted in Fig. 12. For the baseline condition, which is represented as a red line, the deployment time to meet the CAT I VAL presuming that surveying started at the worst possible time of day is approximately two hours. If the multipath condition is worse than the one presented by GAD-A condition, the worst-case CAT I deployment time is approximately 10 hours, as shown by the green line. For the dual frequency (L1/L5) architecture, which is represented as the blue line, the worst-case deployment time to meet CAT-I service is about 0.5 hours. In this case, to compute surveying position, we assumed the use of ionosphere free measurements, and to generate differential corrections, we assumed the use of divergence free smoothing. With this architecture, even for the worst time of day, the deployment time can be reduced down to about 30 minutes for CAT I and even lower for LPV.

As addressed above, the results from feasibility analysis show that at least approximately 0.5 up to 10 hours of system deployment time is required to achieve CAT-I service, depending on different architectural types of mobile DGNSS systems. A further reduction in the deployment time is expected when the proposed fast surveying method is applied. Once the system starts to provide a normal service after obtaining accurate surveyed positions of reference station UAVs, the service area can be extended by moving one of reference station UAVs. Since the other reference stations remain fixed at their sting locations, the one which relocates can almost

immediately estimate its surveyed location by utilizing a RTK technique. More studies pertaining to those topics including experimental validation will be conducted in future.

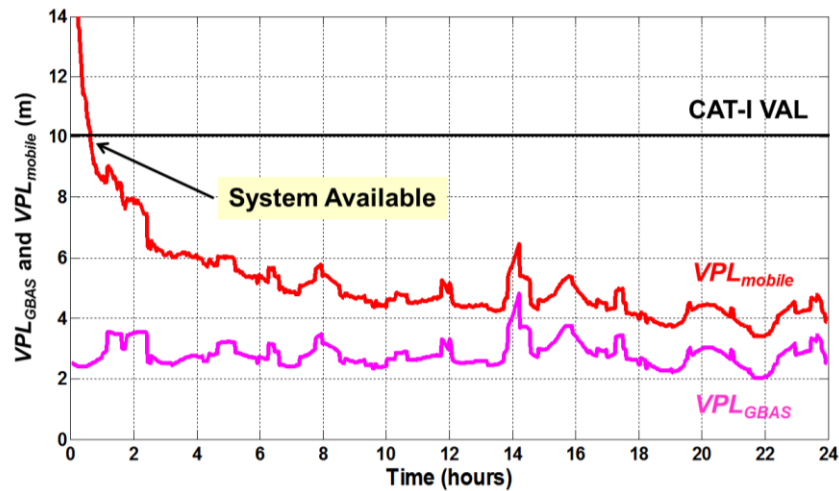


Figure 51 Mobile VPL computed by combining surveying position error with GBAS VPL

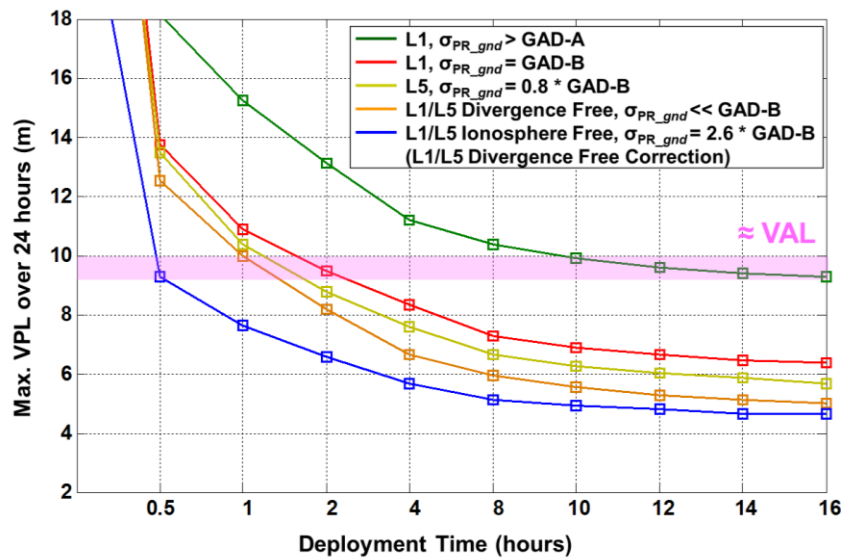


Figure 62 Maximum mobile VPL over 24 hours for different multipath conditions and system architectures and deployment time of mobile DGNSS

6.0 CONCLUSION

This paper presented a concept of mobile DGNSS that is deployable at arbitrary locations in order to provide differential corrections for high-accuracy and high-integrity navigation where needed. This paper introduced preliminary but essential algorithms to address the key issues of mobile DGNSS, which consist of optimal location selection, fast location surveying, and integrity monitoring under non-ideal conditions.

The proposed optimal location selection method allows the user to find appropriate sites for mobile reference stations. In this method, multiple UAVs find point

locations that can fulfill the demands of DGNSS operation. Using image sensors, they land at appropriate locations around the selected points where the terrain is vacant and flat. Next, these reference station UAVs assess their environments using a territorial inclination test and a satellite visibility test. After the UAVs land at their target sites to serve as reference stations, the fast location surveying algorithm is applied to decrease the time taken for the determination of surveyed positions within specified position accuracy requirements. This algorithm specifically requires that relative position measurements between reference stations be very accurate. This paper presents how inter-distance measurements of such accuracy can be obtained from carrier phase

measurements taken from the GNSS receivers. Through simulation tests, the fast location surveying algorithm was verified to be effective in obtaining more accurate positions than the method of simply averaging standalone GNSS position solutions.

The newly designed integrity monitors of mobile DGNSS include a code multipath monitor to detect severe multipath errors and a station displacement monitor to detect possible unexpected movement of the reference stations. The station displacement monitor uses relative positions between the reference stations to verify that the locations of the stations have not moved. The feasibility of the proposed mobile DGNSS concept was assessed by deriving self-surveying error budgets and estimating deployment times for system operation in various ground multipath conditions. In the preliminary assessment, it was found that, without the application of the fast surveying method, at least approximately 0.5 up to 10 hours of system deployment time is required to achieve GBAS CAT-I equivalent service. The assured performance level of the fast location survey algorithm in various realistic environmental conditions will be examined in future works.

The topics discussed in this paper are the major ones since they would have a great influence on the feasibility of the proposed system. However, to realize the proposed conceptual framework into a real system, a continued research should be conducted to solve many other key problems; siting criteria based on multipath error budgets, primary GNSS correction algorithms, communication technologies including datalink and data relay, and survivability skills in hostile environment..

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REFERENCES

- [1] S. Pullen, P. Enge, and J. Lee, "High-Integrity Local-Area Differential GNSS Architectures Optimized to Support Unmanned Aerial Vehicles (UAVs)," *Proceedings of ION ITM 2013*, San Diego, CA, Jan. 28-30, 2013.
- [2] Schultz, A. (2008). *U.S. Patent No. 8165728 B2*. Washington, DC: The United States of America as represented by the Secretary of the Navy.
- [3] D. Lee, M. Kim, K. Kim, and J. Lee, "Fast Location Survey of DGNSS Reference Station to Support UAV Navigation," *ICCAS 2014*, Seoul, October, 2014.
- [4] M-B. Heo and B. Pervan, "Carrier phase navigation architecture for shipboard relative GPS," *IEEE Trans. Aero. and Elect. Sys.*, Vol. 42, No. 2, pp. 670–679,

- 2006.
- [5] P. J. G. Teunissen, "The least-squares ambiguity decorrelation adjustment: a method for fast gps integer ambiguity estimation," *Journal of Geodesy*, Vol. 70, Issue 1-2, pp. 65-82, 1995.
- [6] D. J. Paul, and T. Christian, "The LAMBDA method for integer ambiguity estimation: implementation aspects," *Technical Representative, LGR Series*, No. 12, Delft Geodetic Computing Centre, Delft University of Technology, The Netherlands, 1996.
- [7] Besl, P.J., McKay, Neil D., "A method for registration of 3-D shapes," *Pattern Analysis and Machine Intelligence, IEEE Transactions on*, Vol.14, No.2, pp. 239,256, Feb 1992.
- [8] C. Rocken, C. Meertens, B. Stephens, J. Braun, T. VanHove, S. Perry, O. Ruud, M. McCallum and J. Richardson, UNAVCO Academic Research Infrastructure (ARI) Receiver and Antenna Test Report, 1995.
- [9] G. Xie, "Optimal On-Airport Monitoring of the Integrity of GPS-Based Landing Systems," in *Electrical Engineering*, Stanford University, 2004.
- [10] J. Lee, S. Pullen, S. Datta-Barua and P. Enge, "Assessment of Ionosphere Spatial Decorrelation for Global Positioning System-Based Aircraft Landing Systems," *AIAA Journal of Aircraft*, Vol. 44, No. 5, September-October 2007, pp. 1662-1669.
- [11] "Specification: Performance Type One Local Area Augmentation System Ground Facility," U.S. Federal Aviation Administration, Washington, D.C., FAA-E-2937A, April 17, 2002.
- [12] G. A. McGraw, T. Murphy, M. Brenner, S. Pullen, A. J. Van Dierendonck, "Development of the LAAS Accuracy Models," *Proceedings of the 13th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GPS 2000)*, Salt Lake City, UT, September 2000, pp. 1212-1223.
- [13] Misra, P., and Enge, P., *Global Positioning System: Signals, Measurement, and Performance*, 2nd ed., Ganga-Jamuna, Lincoln, MA, 2006.
- [14] M-S. Ciciu, M. Felux, P. Remi, L. Yi, B. Belabbas, S. Pullen, "Evaluation of Dual Frequency GBAS Performance Using Flight Data," *Proceedings of the 2014 International Technical Meeting of The Institute of Navigation*, San Diego, California, January 2014, pp. 645-656.
- [15] T. Walter, J. Blanch, R. E. Phelts, P. Enge, "Evolving WAAS to Serve L1/L5 Users," *Proceedings of the 24th International Technical Meeting of The Satellite Division of the Institute of Navigation (ION GNSS 2011)*, Portland, OR, September 2011, pp. 2495-2504.
- [16] Minimum Operational Performance Standards for GPS Local Area Augmentation System Airborne Equipment, RTCA, DO-253C, Washington, D. C., Dec. 16, 2008.
- [17] Minimum Operational Performance Standards for Global Positioning System/Wide Area Augmentation System Airborne Equipment, RTCA, Rept. DO-229D, Washington, D. C., Dec. 13, 2006.